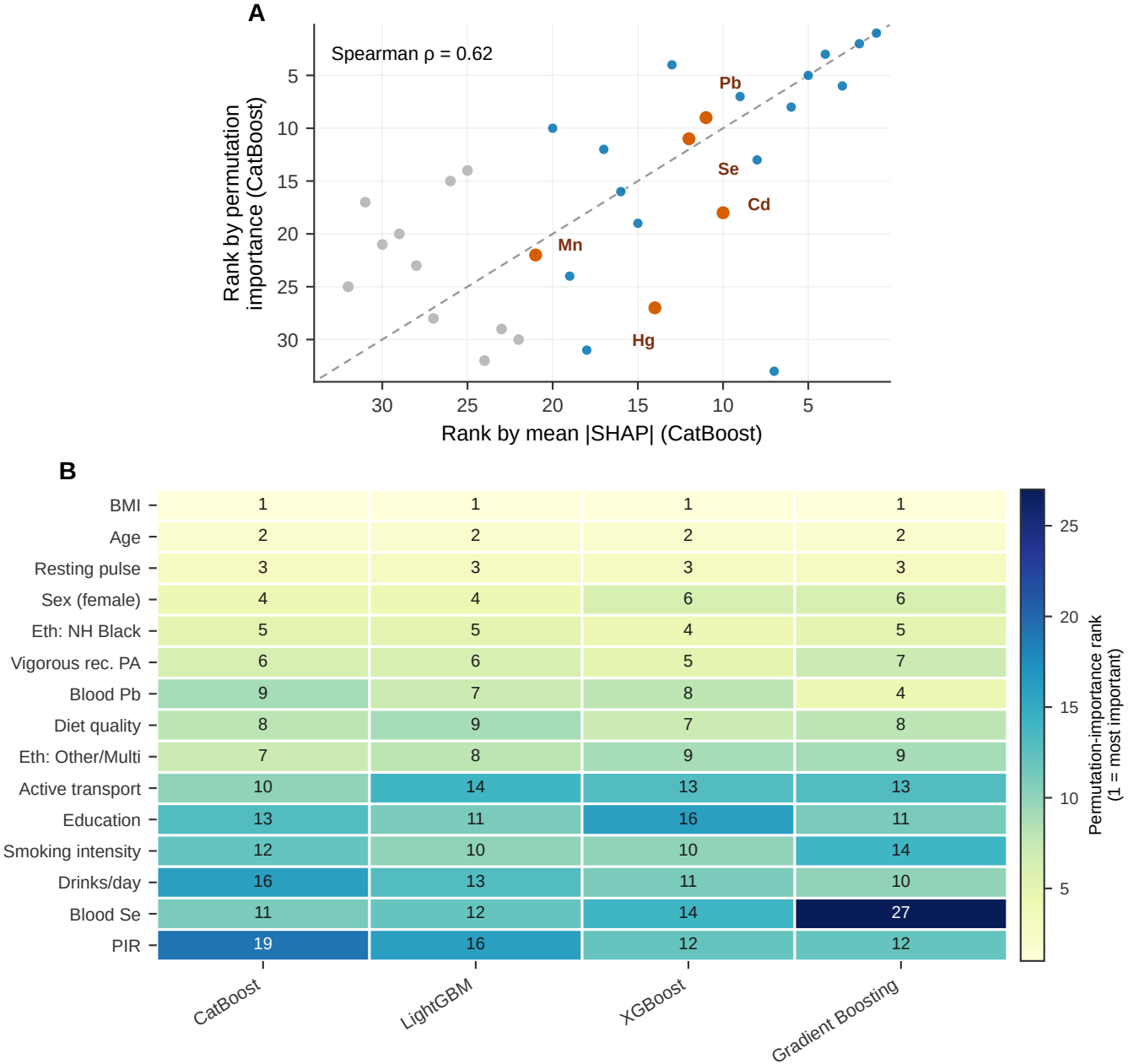
A
Spearman ρ = 0.62
Pb
Se
Cd
Mn
Hg
Rank by permutation importance (CatBoost)
Rank by mean |SHAP| (CatBoost)
B
BMI 1 1 1 1
Age 2 2 2 2
Resting pulse 3 3 3 3
Sex (female) 4 4 6 6
Eth: NH Black 5 5 4 5
Vigorous rec. PA 6 6 5 7
Blood Pb 9 7 8 4
Diet quality 8 9 7 8
Eth: Other/Multi 7 8 9 9
Active transport 10 14 13 13
Education 13 11 16 11
Smoking intensity 12 10 10 14
Drinks/day 16 13 11 10
Blood Se 11 12 14 27
PIR 19 16 12 12
CatBoost
LightGBM
XGBoost
Gradient Boosting
Permutation-importance rank (1 = most important)